Program-09-Documentation

Title: Organize Kubernetes Pods using Labels and Namespaces

Objective: Create isolated environments using Namespaces and tag resources within those environments using Labels to enable efficient management and filtering.

1. Environment Preparation

Initialize the project by creating two primary configuration files:

- nano namespaces.yaml: To define the isolated cluster environments.
- nano pods.yaml: To define the application containers with specific metadata.

2. Defining Logical Isolation (Namespaces)

Namespaces allow you to partition a single Kubernetes cluster into multiple virtual clusters. In this lab, two namespaces were created:

- **dev**: For development workloads.
- **prod**: For production workloads.

```
Terminal
  GNU nano 7.2                    namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

```
Terminal
  GNU nano 7.2                            pods.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:

^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line
```

```
Terminal
  GNU nano 7.2                            pods.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:

^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line
```

3. Resource Tagging (Labels)

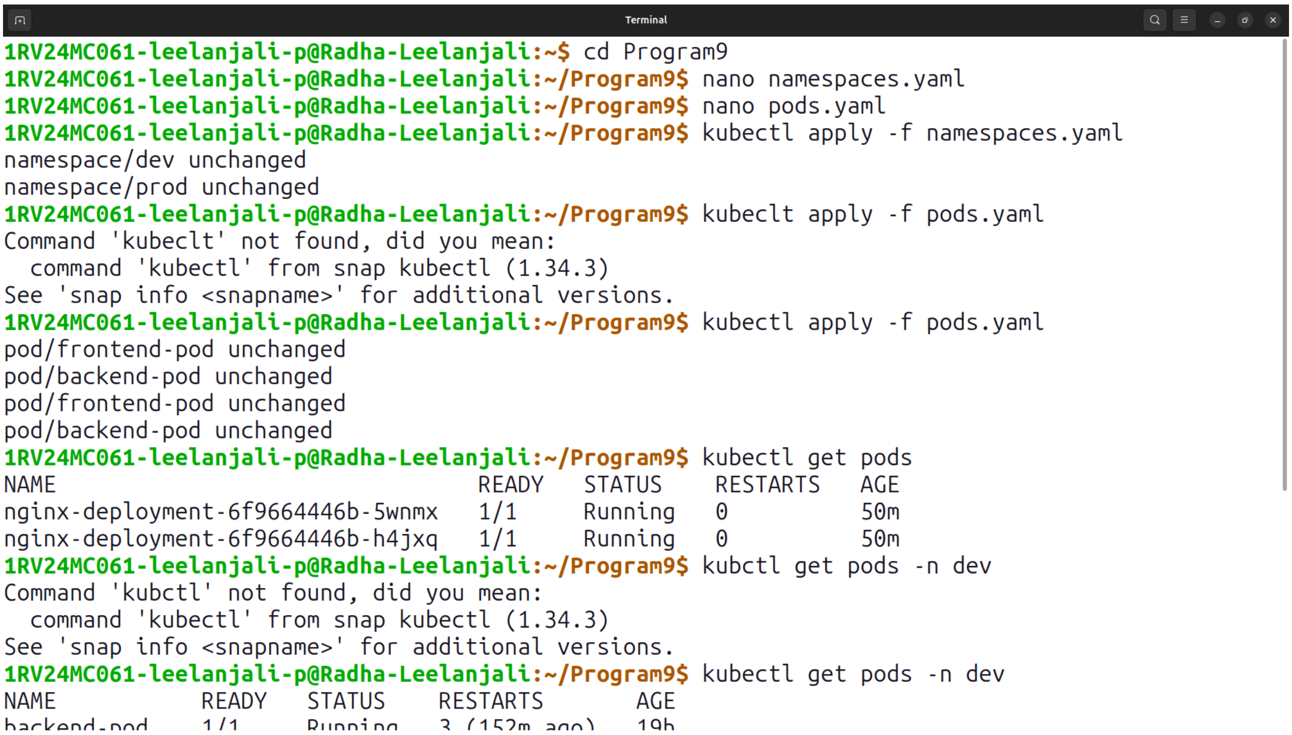
Labels are key-value pairs attached to resources like Pods. They are used to specify identifying attributes that are meaningful to users.

- **Metadata used in pods.yaml**:
    - app: frontend or app: backend (to identify the functional layer).
    - env: dev or env: prod (to identify the environment context).

4. Deployment Execution

The configurations are applied to the cluster using the kubectl apply command.

- **Create Namespaces**: kubectl apply -f namespaces.yaml results in the creation of the dev and prod namespaces.
- **Create Pods**: kubectl apply -f pods.yaml deploys the specified pods (e.g., frontend-pod and backend-pod) into their respective namespaces.



```
Terminal
1RV24MC061-leelanjali-p@Radha-Leelanjali:~$ cd Program9
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ nano namespaces.yaml
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ nano pods.yaml
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ kubectl apply -f namespaces.yaml
namespace/dev unchanged
namespace/prod unchanged
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ kubeclt apply -f pods.yaml
Command 'kubeclt' not found, did you mean:
  command 'kubectl' from snap kubectl (1.34.3)
See 'snap info <snapname>' for additional versions.
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ kubectl apply -f pods.yaml
pod/frontend-pod unchanged
pod/backend-pod unchanged
pod/frontend-pod unchanged
pod/backend-pod unchanged
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-5wnmx   1/1     Running   0          50m
nginx-deployment-6f9664446b-h4jxq   1/1     Running   0          50m
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ kubctl get pods -n dev
Command 'kubctl' not found, did you mean:
  command 'kubectl' from snap kubectl (1.34.3)
See 'snap info <snapname>' for additional versions.
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ kubectl get pods -n dev
NAME            READY   STATUS    RESTARTS       AGE
backend-pod     1/1     Running   3 (152m ago)   19h
```

5. Advanced Resource Filtering

Once resources are deployed, you can use **Selectors** (the -l flag) and **Namespace flags** (the -n flag) to filter the output.

Action	Command Used	Result
List All Pods (Default)	kubectl get pods	Shows pods in the default namespace only (e.g., nginx-deployment).
List Pods in Namespace	kubectl get pods -n dev	Shows only the frontend-pod and backend-pod assigned to the dev environment.
Filter by Label	kubectl get pods -n dev -l app=frontend	Precisely targets and lists only the frontend component within the dev namespace.
Verify Production	kubectl get pods -n prod	Confirms that a separate set of pods is running in the production namespace.

Terminal

```
pod/frontend-pod unchanged
pod/backend-pod unchanged
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-5wnmx   1/1     Running   0          50m
nginx-deployment-6f9664446b-h4jxq   1/1     Running   0          50m
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ kubctl get pods -n dev
Command 'kubctl' not found, did you mean:
  command 'kubectl' from snap kubectl (1.34.3)
See 'snap info <snapname>' for additional versions.
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ kubectl get pods -n dev
NAME           READY   STATUS    RESTARTS       AGE
backend-pod    1/1     Running   3 (152m ago)   19h
frontend-pod   1/1     Running   3 (152m ago)   19h
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ kubectl get pods -n prod
NAME           READY   STATUS    RESTARTS        AGE
backend-pod    1/1     Running   3 (152m ago)    19h
frontend-pod   1/1     Running   2 (5h10m ago)   19h
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ kubectl get pods -n dev -l app=frontend
NAME           READY   STATUS    RESTARTS       AGE
frontend-pod   1/1     Running   3 (153m ago)   19h
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$ kubectl get pods -n dev -l app=backend
NAME          READY   STATUS    RESTARTS       AGE
backend-pod   1/1     Running   3 (153m ago)   19h
1RV24MC061-leelanjali-p@Radha-Leelanjali:~/Program9$
```